Microphones and loudspeakers

A microphone creates an electrical wave called an audio signal. This electrical wave is a copy of the variations in air pressure of an incoming sound wave. When the audio signal is amplified, or strengthened, and fed through a loudspeaker, the original sound is reproduced and can be increased in volume.

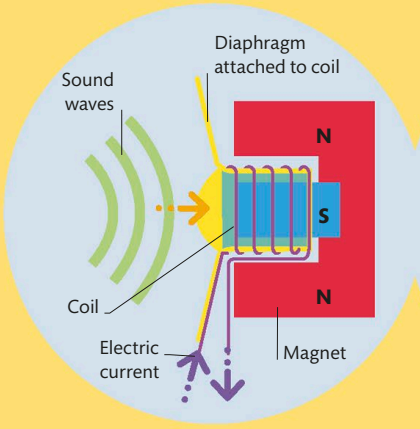
SHOULD I WEAR EARPLUGS TO A CONCERT?

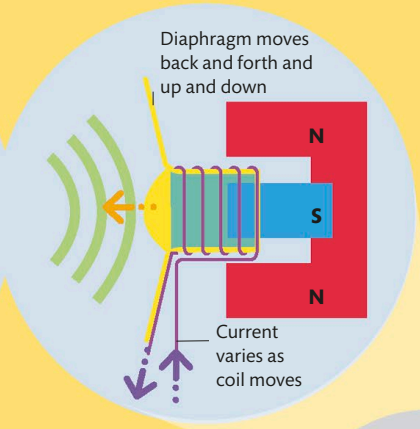
The loudspeakers at a pop concert can produce huge variations in air pressure that can damage your ear, so ear plugs are a good idea if you are close to them.

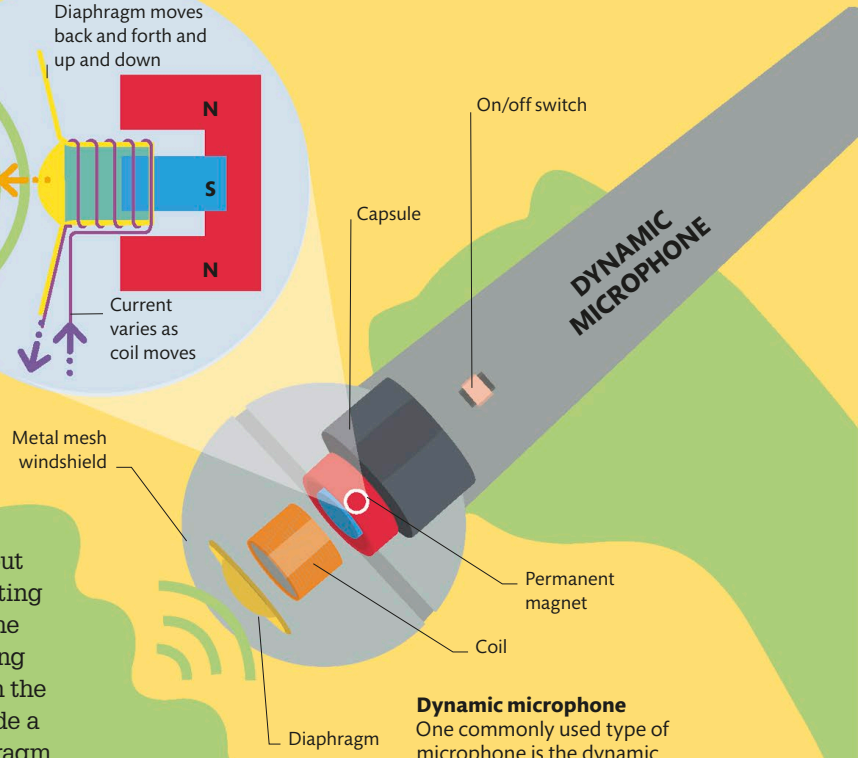
1 Diaphragm in
As a sound wave travels into the microphone, it passes through a layer of protective metal mesh before reaching the diaphragm, which is connected to a thin wire coil. High-pressure air pushes the diaphragm inward, moving the coil down.

2 Diaphragm out
Low-pressure air allows the diaphragm to move back again. As a result, the diaphragm moves to and fro with the rapid variations in pressure of any sound waves that hit it. As the diaphragm moves inward and outward, it takes the coil of thin wire with it.

3 Audio signal generated
The coil surrounds one pole of a permanent magnet, and the movement creates an electric current that flows first one way and then the other. This alternating current, the audio signal, is a copy of the variations of pressure in the sound wave.







Catching sound waves

Sound is a disturbance of the air that travels out from its source in the form of waves of alternating high and low air pressure (see pp.136–137). The audio signal a microphone produces is a varying electric current: the variations in current match the variations of pressure in the sound wave. Inside a microphone is a thin membrane called a diaphragm. The sound waves make the diaphragm vibrate to and fro when they hit it—it is this movement of the diaphragm that creates the electrical signal.

Dynamic microphone
One commonly used type of microphone is the dynamic microphone. Inside it, the diaphragm vibrates a coil of wire positioned around a magnet, producing an alternating electrical current.